Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



TEAM DETAILS · AI BUILDER CUP 2026 · JAPAC · SUSTAINABILITY

Team: Agent Ram · Leader: Ramasamy P · Product: TerraSentinel AI

Satellites leave 1,355 moving points. The engine ranks the few that matter.

Gemini writes the ticket — every number checked. Live on Cloud Run + BigQuery.

Problem: InSAR portals give points, not who goes today. Chatbots invent mm.

Solution: 250 m sites · RPI · honest ETA bands · Gemini number guard.

TERRASENTINEL AI

Satellites see the ground move.

Gemini tells engineers where to go first.

62,330 → 4

Observations become act-today sites

RPI 86 · CRITICAL

NH-58 · 35–64 day outlook if trend continues

Gemini · verified

6 tools · 7 numbers checked · 24h dispatch



PROBLEM & DECISION

Ground creeps for months. Nobody ranks which creep matters — crews go to the wrong place.

NH-58 mountain highway
Sole valley access. One slip cuts towns, pilgrims, supply.

Chennai metro tunnels
Settlement under roads & buildings. Tight tolerances.

Dams & tailings
Crest/toe motion · severe downstream consequence.

The one decision
After this pass: which 250 m, why, and how soon?

Today fails the decision

- Raw InSAR portal — thousands of points, no asset link, no who-goes-today
- Rainfall alarm — treats the trigger as the hazard; fires on stable ground
- Generic AI chatbot — invents mm and failure dates; nothing to audit
- Scheduled walk-over — calendar ignores which segment is accelerating now

We do
- Progressive creep on linear assets
- Rank · explain · track between passes
- Threshold window + assumptions stated

We refuse
- Failure / collapse dates
- Sudden events between 6–12 d passes
- Certified safety decisions

One pass · demo data

62,330	1,355	87	7	4
observations	radar points	250 m sites	CRIT/ELEV	Act today

WHY WE WIN

Not another InSAR viewer — a consequence-ranked decision layer with guarded GenAI.

Differentiation

- 250 m chainage sites on corridor geometry — not a heat-map of every scatterer
- RPI = hazard × consequence × confidence (deterministic Python / BigQuery)
- Act list = today's state; Alerts = what changed between passes
- Rain amplifies only (+50% cap); USGS quakes never move a score
- Gemini narrates tool results; the engine owns every number
- WEAK radar guard — bad data can never become CRITICAL

How it solves it

- Validate → locate on assets → measure speed / accel / progress
- Classify with data-quality gate; rank Act-today shortlist
- Outlook = [ETA_const, ETA_accel] band ≤ 365 d — never a date
- Work order: summary · hypothesis · urgency · field checklist
- Same maths locally or in BigQuery (TS_DATA_BACKEND flag)
- History is re-run from data — every past score is reproducible

USP · live proof

- Honest continuing-regime bands with assumptions spelled out
- Number guard rejects invented mm and "will collapse" language
- Live Cloud Run + BigQuery + Secret Manager + GCS state
- Demo funnel: 62,330 obs → 87 sites → 7 action → 4 Act today
- Gemini live: 6 tool calls · 7 numbers verified · 24h dispatch
- Judge can sign in now — credentials on the login page

PRODUCT SURFACES

Every screen ships a decision an engineer can act on today.

Command Center

Map + KPIs + Act today — ranked by consequence, not raw mm

Site drawer

RPI, speed vs baseline, rain trigger, outlook band, pass trend

Alerts

Pass-to-pass diffs — what changed, not a duplicate Act list

Trends · Portfolio

16 passes recomputed from data; action sites 3→7 tracked

Work Orders

Template-first ticket; Gemini polish; number guard on every figure

Data Hub

Validate-then-import · Open-Meteo rain · provenance · BQ sync

Access & audit

admin/analyst/viewer · CSRF · sessions · full audit log

Methodology

Published formulae — every score auditable by a geotech

STORY & PROCESS FLOW

One satellite pass — from space to a crew at the right 250 metres.

1 · Pass
Every 6–12 days

Sentinel-1 measures mm motion through cloud & night

2 · Map
Onto the corridor

1,355 points matched to highway / metro / dam → 87×250 m sites

3 · Rank
What matters

Fast + accelerating + critical + good data rises; rain alone never alarms

4 · Gemini
Write the ticket

7 tools → reason → JSON → number guard → field checklist

5 · Next
Keep watching

Crew acts; next pass shows calm-down or worse

Demo funnel (labelled synthetic)

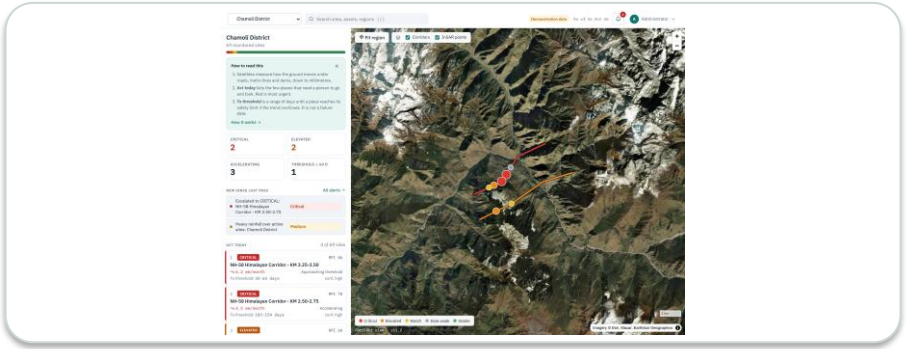
62,330	1,355	87	7	4
obs	pts	sites	action	Act today

Who it serves

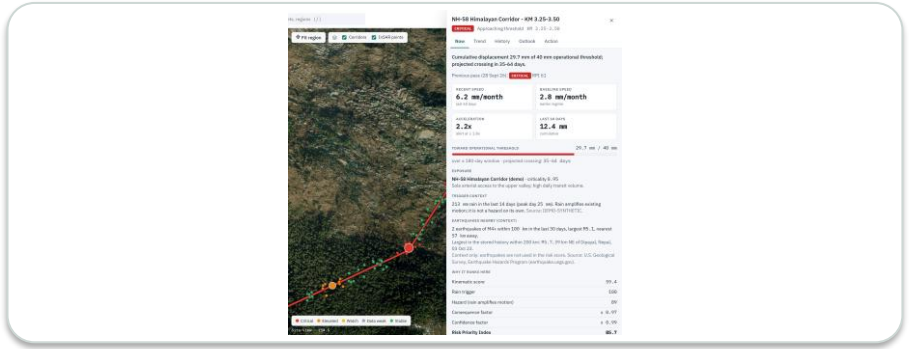
- Public: find problems while small — keep roads open
- Engineers: short list + ticket (what / where / why)
- Leaders: spend crews where risk is rising

PRODUCT UI (LIVE)

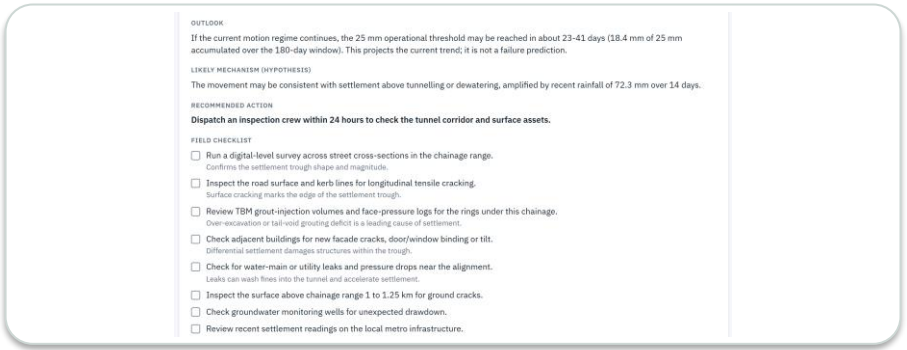
What judges open today — Command Center → site → Gemini ticket → Data Hub.



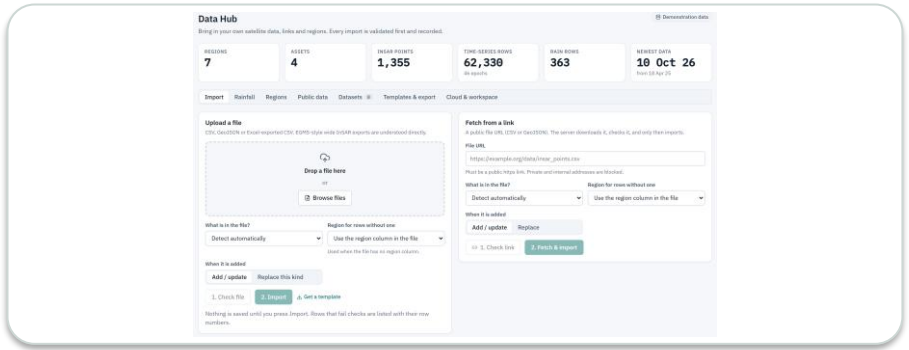
Command Center · Act list



Site · RPI & outlook



Gemini work order



Data Hub · validate then import

ARCHITECTURE STORY

Ingest → Store → Compute → Snapshot → Serve. One container. GenAI never owns the maths.

Ingest	Store	Compute	Snapshot	Serve
Data Hub validates files & HTTPS links	BigQuery or local GIS join · history	Detection engine RPI · tier · band	Pass replay alerts · trends	UI + Gemini ticket audit trail

EDGE	Cloud Run · HTTPS · CSP/HSTS · rate limits · sessions · GCS state
UI + API	React/MapLibre · FastAPI · RBAC · CSRF · SSRF-safe fetch · audit
ENGINE	Deterministic RPI / tiers / ETA bands / alerts — GenAI never computes these
GENAI	Gemini tools → number guard → template fallback (Vertex or AI Studio)
DATA	BigQuery partitions/GEOGRAPHY XOR local · USGS context only · provenance

GENAI ON GOOGLE CLOUD

Without GenAI: a number. With Gemini + guard: a ticket an engineer can trust.

LIVE PROOF: Gemini drafted a Metro Line 4 ticket with 6 tool calls and 7 numbers verified. Recommendation: “Dispatch an inspection crew within 24 hours…” — every millimetre from tools, not free text.

Ingest	Store	Compute	Explain	Act
Cloud Run Data Hub	BigQuery GIS + history	Engine RPI · tier · band	Gemini + number guard	Map · ticket · audit

Gemini loop

- 7 tools: metrics, history, outlook, rain, seismic, exposure, quality
- Round 0 MUST call tools → investigate → write JSON
- Temp 0.1 · model fallback on 429/5xx

Number guard

- Rejects any mm/days/% not in a tool result
- Rejects “will collapse”, “failure date”, “no risk”
- On failure: labelled template still ships

GCP stack

- Cloud Run — SPA + API + engine + agent
- BigQuery — partitions, ST_DWITHIN, bytes cap
- Vertex/AI Studio · Secrets · GCS · ADC

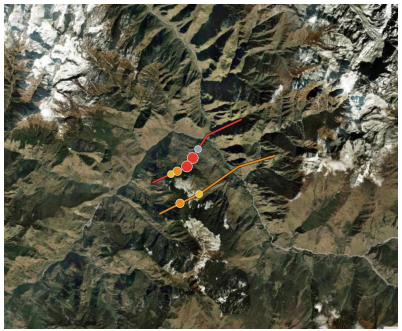
ESTIMATED COST (OPTIONAL)

GenAI only on tickets; ranking stays deterministic CPU.

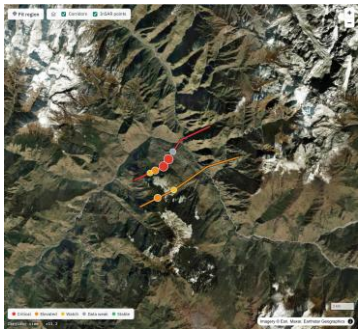
Cloud Run	1–2 vCPU / 2 GiB · min=1 while judging	low tens USD/mo warm
BigQuery	~62k rows · TTL cache · max bytes billed	pennies–low USD demos
Gemini Flash	Work-order path only · template fallback	per-ticket, optional
Secrets + GCS	Admin/Gemini secrets · state volume	pennies
Scale path	Materialize assignment · raise instances	linear with corridors

PROTOTYPE SNAPSHOTS

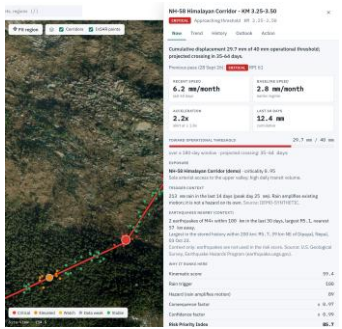
Winning proof on screen — Chamoli Act list, NH-58 CRITICAL, verified Gemini ticket.



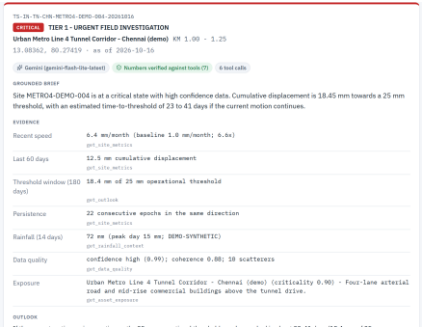
Chamoli · 69 sites · 4 act today



Ranked map — not every scatterer



NH-58 · RPI 86 · 35-64 day band



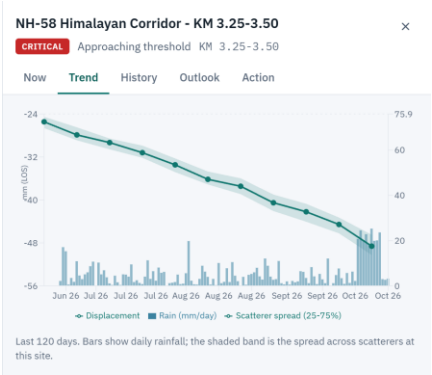
Gemini · numbers verified

WORKED PROOF

NH-58 KM 3.25–3.50 — same numbers in UI, tools, and Gemini work order.

CRITICAL · RPI 86 · band 35–64 d

- Inputs: 6.2 vs 2.8 mm/mo · 2.2× · 29.7/40 mm · rain 213 mm · crit 0.95 · conf 0.97
- kin = 100×(.4·0.62 + .3·0.40 + .3·0.74) = 59.4
- hazard = 59.4×(1+0.5×1.0) = 89.1 (rain capped)
- RPI = 89.1×0.975×0.985 ≈ 85.7 → CRITICAL (threshold approach)
- Outlook: 51 d const · 44 d accel → widened 35–64 d if trend continues
- History: this site 61→86 · portfolio action sites 3→7 over 16 passes
- What-if: minor road→57 ELEV · weak radar→≤15 WEAK · rain on stable→≈1
- Integrity: GenAI never computes RPI / tier / ETA — tools only



Site trend · displacement + rain

Live Cloud Run + BigQuery (warm)

summary ~0.4s · sites ~0.3s · alerts ~2s

Hot-path prefetch before history warm

Gemini: 6 tool calls · 7 numbers verified

ASK · BUYER · ROADMAP

Sustainability impact today. Scale tomorrow. Honesty rules never move.

THE ASK: Review the live demo, the 3-minute video, and the repository.

Buyer: asset-integrity / geotech teams. They buy engineer attention — 1,355 points become 7 sites.

Roadmap

- Production InSAR providers on the same Data Hub schema
- Materialize spatial assignment for national corridors
- HA when session store leaves single-process files
- OIDC / IAP · field offline packet · deeper amplifiers only

Non-negotiable honesty

- Outlook bands ≠ collapse / failure dates — always
- Gemini narrates tool numbers only — always
- Synthetic demo data labelled in the UI — always
- Secrets in Secret Manager / .env — never in git

JUDGE PATH · LINKS

Sign in → Command Center → NH-58 site → Work Order → Data Hub (5 minutes).

1 · GitHub

https://github.com/ramasamyp-aiengineer/terra-sentinel-ai

2 · Demo video (3 min)

docs/pitch/TerraSentinel-AI-Demo-3min.mp4 — upload to Drive/YouTube for the portal

3 · Live product

https://terrasentinel-598899050820.asia-southeast1.run.app

Credentials are shown on the sign-in page (judge demo hint).

Look for: Act list · outlook band (not a date) · “numbers verified” on the work order.

RISK ENGINE MATHS

Measure → Score → Project. Deterministic, tested, explainable.

1 · Measure

- Recent speed: slope last 60 d
- Baseline: 270 to 90 d ago
- Accel = |recent|/max(|base|,0.5)
- Persistence · seasonal recovery
- Site = median in 250 m bin
- conf mixes coh·epochs·pts·SNR
- Weak data can never be CRITICAL

2 · Score

- kin=100*(.4s+.3a+.3p)
- hazard=kin*(1+0.5*rain/200)
- RPI=hazard*(0.5+0.5*crit)
- *(0.5+0.5*conf)
- Slow/far x0.2 · seasonal x0.35
- WEAK cap 15 · quakes never scored
- CRITICAL / ELEVATED / WATCH

3 · Project

- ETA_const = remaining / speed
- ETA_accel = quadratic root
- band=[min,max] <= 365 d
- Highway 40 mm / 180 d demo
- Metro 25 mm · Dam 30 mm
- ALWAYS a band
- NEVER a failure date

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

